FHIA-05 - AI in the Healthcare Information Architecture

Transforming Data into Information, Knowledge, and Insight

DOUGLAS EBEL

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AI in healthcare is often portrayed as taking patient information and producing a diagnosis or treatment recommendation. FHIA-05 proposes a different role: using AI to help manage the growing volume and complexity of healthcare information. It examines how AI can extract clinical information from healthcare data, reconcile it into an evidence-linked Longitudinal Patient History, and analyze authorized knowledge to produce useful insight for providers, payers, patients, and other healthcare purposes—while preserving the evidence needed for human decisions.

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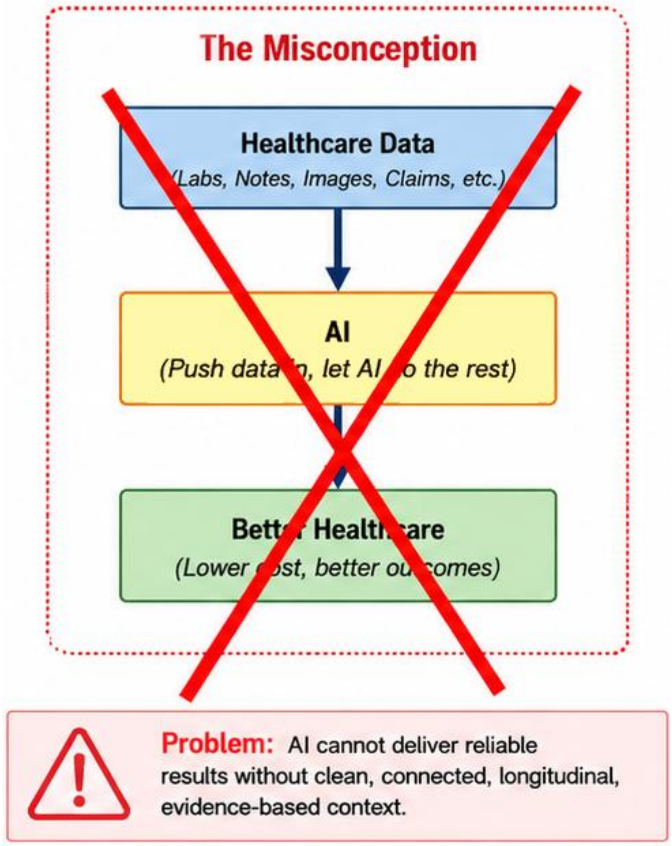
Introduction

A common vision for AI in healthcare is deceptively simple: give an AI system everything known about a patient and ask it what is wrong and what treatment or medication to prescribe. That vision focuses attention on replacing or automating clinical judgment. It overlooks a more immediate problem: healthcare already generates more information than the people making healthcare decisions have the time to find, reconcile, analyze, and understand.



A national longitudinal patient history makes far more of that information available, but availability alone does not make it useful. Decades of clinical notes, diagnoses, laboratory results, imaging, medications, procedures, treatments, claims, and outcomes cannot simply be placed in front of a physician during a fifteen- or twenty-minute encounter. Nor can every payer, pharmacist, specialist, or other healthcare participant be expected to manually reconstruct the portions of that history relevant to each decision.

The role of AI proposed in the Future Healthcare Information Architecture is therefore not simply **information in, medical decision out**. AI is used throughout the architecture to help transform **Data → Information → Knowledge → Insight**, providing the foundation for a Human Decision. It can extract clinical information from healthcare data, reconcile that information across sources and time, identify what is significant for a particular healthcare purpose, and present the resulting insight to the person responsible for making the decision.

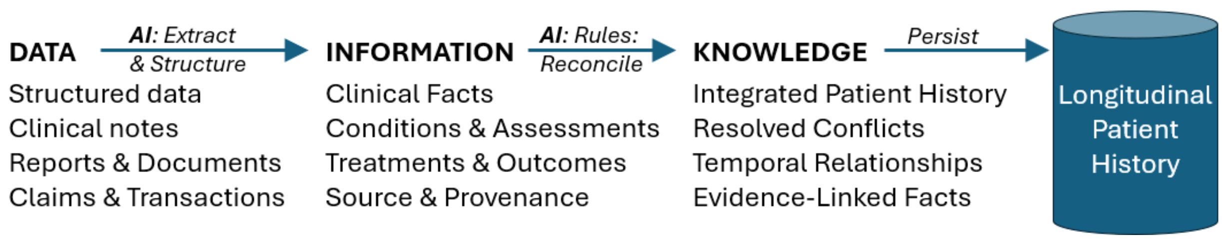
The objective is not to ask AI to replace healthcare professionals. It is to use AI to perform information-processing work that increasingly exceeds the time humans can reasonably devote to it—while preserving the evidence they need to understand, verify, and ultimately decide what to do.

Building and Using Longitudinal Knowledge

From Healthcare Data to Longitudinal Knowledge

Every healthcare encounter creates new data. Some is captured as structured data in Electronic Health Records, laboratory, imaging, pharmacy, payer, and other Healthcare Information Systems. Other information is contained in clinical notes, reports, forms, and documents written primarily for people rather than computers. The Future Healthcare Information Architecture uses both forms to build and maintain the patient's Longitudinal Patient History continuously.

After Each Encounter



The first two transformations occur after new healthcare data becomes available. Their purpose is not to make a clinical decision, but to make what happened during the encounter understandable in the context of everything previously known about the patient.

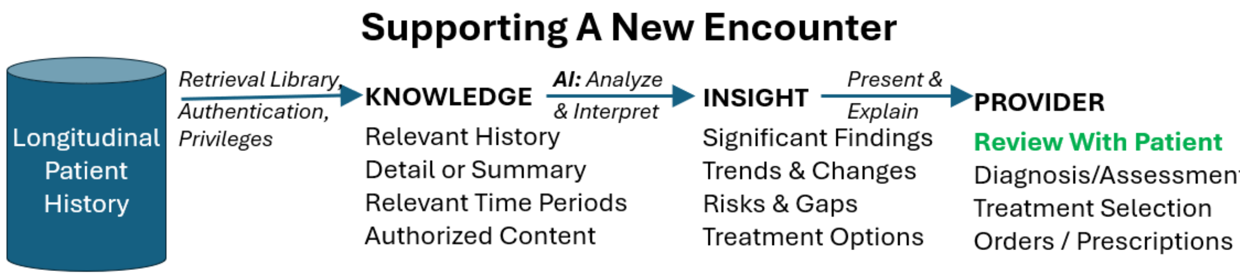
Data → Information. Structured data can often be interpreted directly, while AI can extract clinically meaningful facts from unstructured clinical notes and reports. Conditions, assessments, treatments, outcomes, and other facts are identified and structured while preserving their source and provenance. The objective is to capture what each source says without losing the evidence from which it was derived.

Information → Knowledge. New information is then integrated with information accumulated across previous encounters and healthcare organizations. Repeated information can be consolidated, conflicting information identified and reconciled, and relationships across time established. The result is an evidence-linked understanding of the patient's history that is persisted in the Longitudinal Patient History while retaining the underlying information needed to verify how that knowledge was derived.

These transformations require AI and deterministic processing to perform specific tasks consistently according to defined rules. We discuss how those capabilities are developed, constrained, and validated later.

From Longitudinal Knowledge to Insight

The Longitudinal Patient History may eventually contain decades of knowledge assembled from encounters across many healthcare organizations. Making all that knowledge available does not mean presenting it for every purpose. A physician preparing to see a patient, a payer reviewing a prior authorization, and an analyst evaluating treatment effectiveness require different portions of the same longitudinal history.



A library of standardized retrieval requests can provide detailed or summarized views of the Longitudinal Patient History for different purposes. Authentication identifies who is making the request, while privileges determine what that individual or organization is permitted to retrieve. The result is a purpose-specific set of longitudinal knowledge rather than the patient's entire history.

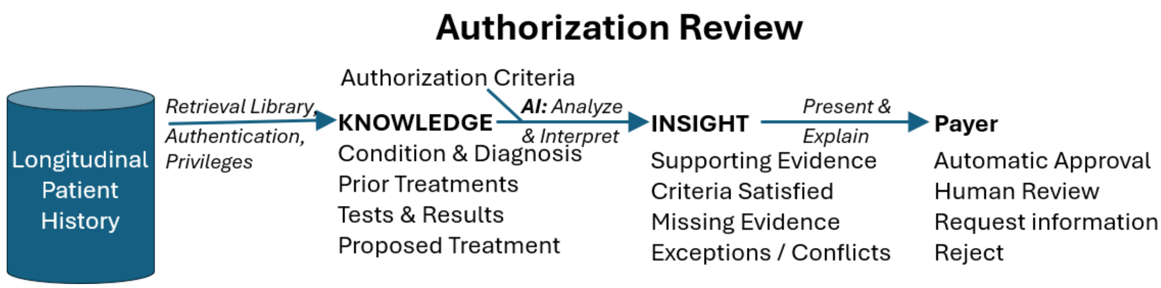
Knowledge → Insight. AI analyzes the retrieved patient knowledge, together with relevant external knowledge when required, to determine what is significant for the particular purpose. For a provider encounter, this might include important findings from the patient's history, trends and changes over time, risks or gaps requiring attention, and possible treatment options informed by clinical guidelines or evidence. The objective is not simply to summarize what was retrieved, but to identify what deserves the provider's attention and preserve the evidence supporting those observations.

Insight → Human Decision. The resulting insights are presented to the provider before or during the encounter. The provider can examine the supporting evidence, consider information obtained directly from the patient, discuss alternatives with the patient, and decide what action is appropriate. AI reduces the information-processing burden; responsibility for the clinical decision remains with the healthcare professional.

One History, Multiple Uses

The same Longitudinal Patient History can support many different participants and purposes without creating a separate patient history for each application. Providers, payers, patients, researchers, public health organizations, and government agencies may draw upon the same underlying longitudinal knowledge. Still, they do not need the same information or use it in the same way.

The retrieval library provides purpose-specific views of the history, while authentication and privileges control what each requester may access. AI can then analyze the retrieved patient knowledge for the particular task, combining it when necessary with external knowledge, rules, or criteria appropriate to that use. The resulting insight is therefore specific to both the patient and the purpose for which the knowledge was requested.



Prior authorization provides one example. A payer could retrieve the portions of the patient's history relevant to a requested treatment and compare that evidence with its authorization criteria. Rather than requiring provider and payer personnel to assemble and review supporting documentation repeatedly, AI could identify which requirements are supported by the longitudinal history, where the supporting evidence resides, and what information may still be missing. The payer's authorization criteria remain separate from the patient's history, while the Longitudinal Patient History provides the evidence needed to apply them.

Examples of How Longitudinal Knowledge Can Be Used

The Longitudinal Patient History can support many healthcare questions using the same basic process: retrieve the knowledge appropriate to the purpose, analyze it to produce insight, and return only the information the requester is authorized to receive. The examples below illustrate how the retrieved knowledge and AI analysis change with the purpose, as well as whether the resulting insight requires an identified patient, de-identified patient histories, or only summary information.

From Healthcare Questions to Actionable Insight

Purpose / Question	Knowledge Retrieved	AI Analysis	Insight / Privacy
What treatments have already been tried?	Condition, treatments, dates, response, side effects, outcomes	Relate treatments and outcomes across encounters; identify unsuccessful or discontinued treatments	**Patient identified:** treatment history relevant to the current condition
How have test results changed over time?	Test type, results, reference ranges, dates, related conditions and treatments	Normalize comparable results; identify trends, significant changes and anomalies	**Patient identified:** trends and changes with supporting results
Which treatment works best for patients like this?	Condition, patient characteristics, treatments, outcomes, complications	Find comparable patients; compare outcomes among treatment alternatives	**Patient de-identified:** treatment effectiveness summarized for comparable patients
Which provider has the most experience—and best outcomes?	Conditions treated, procedures/treatments, patient risk factors, outcomes, provider	Risk-adjust provider experience and outcomes for comparable patients	**Patients de-identified; providers identified:** provider-level comparative results
Should this test be approved?	Condition, symptoms, prior tests, treatments, outcomes, proposed test	Compare patient evidence with payer authorization criteria	**Patient identified:** criteria satisfied, supporting evidence and missing requirements
Which payer policies cause harmful delays?	Authorization requests, criteria, approvals/denials, treatment timing, subsequent outcomes	Compare delays and outcomes associated with authorization policies	**Patients de-identified:** summary results by policy/payer

From Healthcare Questions to Actionable Insight

Purpose / Question	Knowledge Retrieved	AI Analysis	Insight / Privacy
Is this drug causing unexpected problems?	Medication, dosage, duration, patient characteristics, conditions, adverse outcomes	Identify outcome patterns associated with the drug and potentially affected populations	**Patients de-identified:** aggregate adverse-outcome patterns
Where are providers needed?	Patient demographics/geography, conditions, referrals, provider specialty, travel/access patterns	Identify geographic demand, shortages and access gaps	**Summary data:** geographic need without patient identification
Where is healthcare fraud occurring?	Claims, diagnoses, procedures, treatments, providers, clinical evidence, utilization patterns	Identify inconsistencies and unusual patterns requiring investigation	**Initially de-identified/summary:** identified records available only for authorized investigation
What conditions run in the family?	Patient conditions plus authorized family relationships and relevant family histories	Identify recurring conditions and patterns among related individuals	**Patient identified:** relevant familial risks; relatives disclosed only as permitted

How AI Is Used in the Architecture

The preceding examples illustrate that AI is not a single step between healthcare data and a decision. It performs different information-processing tasks at different points in the architecture. After an encounter, AI helps extract and structure clinical information from newly received data and reconcile that information with the patient's existing longitudinal history. When that history is later used, AI analyzes purpose-specific knowledge to identify the findings, relationships, trends, risks, or options that deserve attention. These three uses of AI—**Extract & Structure, Reconcile & Integrate, and Analyze & Interpret**—require different instructions, rules, knowledge, and methods for determining whether the results are reliable.

Extract & Structure: Data → Information

Healthcare data arrives in many forms. Some information is already structured—laboratory results, prescriptions, procedure codes, claims, vital signs, and other discrete fields can often be mapped directly into the longitudinal data model. Much of the clinical meaning, however, remains embedded in physician notes, imaging reports, discharge summaries, referral documents, and other narrative text. AI, particularly Large Language Models (LLMs), can help transform this unstructured data into structured clinical information.

The task is more difficult than identifying medical terms in text. A clinical note may mention a condition because the physician diagnosed it, the patient reported having it years ago, a family member had it, it is being considered as a possible diagnosis, or the physician specifically ruled it out. Similarly, a treatment may have been performed, recommended, considered, discontinued, or reported by the patient as having occurred elsewhere. Extracting a term without understanding its context could turn what the note actually says into incorrect patient information.

Patient reports right shoulder pain for six weeks. A rotator cuff injury 10 years ago was treated with physical therapy. Examination shows limited range of motion and pain with abduction. Possible rotator cuff tendinopathy; a tear cannot be excluded. Acetaminophen has provided limited relief. Physical therapy is recommended. MRI will be considered if symptoms persist

Legend

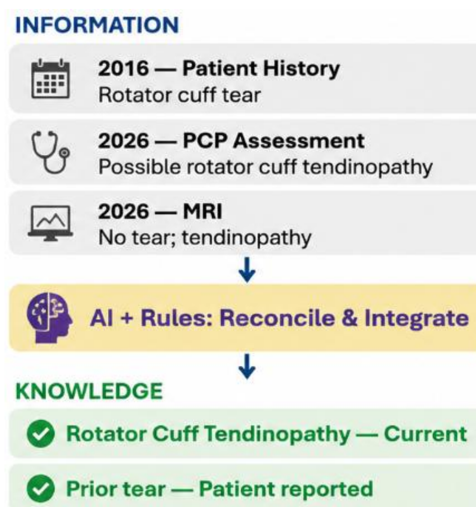
Condition/Finding
Past Treatment
Future Treatment

The AI must therefore be taught what information to extract and how to classify what it finds. For each relevant statement, it may need to identify the clinical concept, its status, when it occurred, how it relates to the current encounter, and whether it represents a physician assessment, patient statement, test result, treatment, outcome, or other type of evidence. The output should conform to the defined longitudinal data model rather than allowing the AI to invent its own representation.

Just as important, the transformation must preserve provenance. An extracted fact should remain linked to the note, report, or structured record from which it was derived. The objective is not merely to create more structured data; it is to create clinical information whose meaning and source can later be evaluated, reconciled, and, when necessary, verified against the original evidence. When an extraction is found to be incorrect, that correction can also serve as feedback to improve the extraction process and reduce similar errors in the future.

Reconcile & Integrate: Information → Knowledge

Extracting information from each healthcare encounter does not by itself create a longitudinal understanding of the patient. The same condition may be described differently by multiple providers, appear repeatedly over many years, change in severity or status, or be contradicted by later information. Treatments may occur across different organizations, and their outcomes may not be documented until months or years later. The architecture must relate each new piece of information to what is already known about the patient.



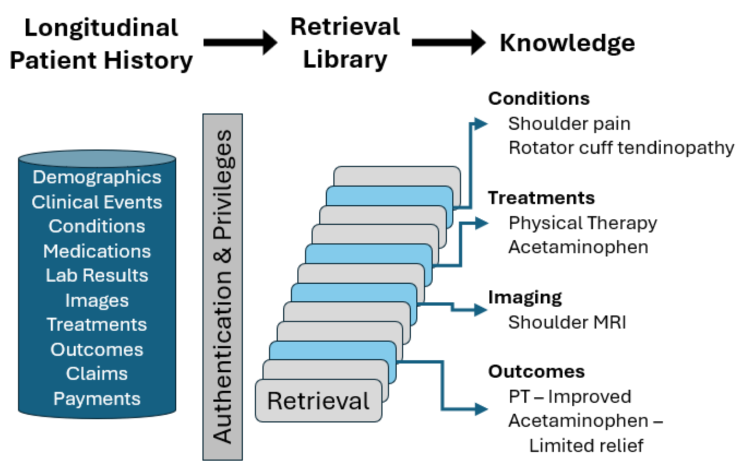
AI can help determine whether newly received information confirms, updates, supplements, or conflicts with existing information. A new diagnosis may establish a previously unknown condition, confirm an existing condition, refine its description, indicate that it has resolved, or contradict an earlier assessment. Similarly, a treatment must be related to the condition it was intended to address, and subsequent results, complications, or side effects must be associated with that treatment when the available evidence supports the relationship.

Reconciliation cannot simply assume that the newest information is correct. A patient's answer on a current intake form may conflict with years of clinical evidence, while a recent diagnostic test may legitimately overturn an earlier diagnosis. AI must therefore apply defined rules for considering the source, type, timing, and strength of the available evidence. Where the evidence does not support a definitive conclusion, the uncertainty or conflict should remain visible rather than being resolved arbitrarily.

The result is knowledge that represents what the accumulated evidence establishes about the patient across organizations and time. The original information and its provenance remain available, allowing a derived conclusion to be traced back to its supporting or conflicting evidence. In this way, the Longitudinal Patient History becomes more than a collection of records: it maintains an evidence-linked understanding of conditions, treatments, outcomes, and their relationships as that understanding changes over time.

Retrieving Knowledge for a Purpose

Before AI can analyze knowledge from the Longitudinal Patient History, it must retrieve the knowledge appropriate to the purpose. Rather than giving an AI system unrestricted access to everything known about a patient, the architecture can provide a library of standardized retrieval services designed for different healthcare uses. A request might retrieve conditions and their history, treatments and outcomes, medications, laboratory trends, prior procedures, or other defined combinations of longitudinal knowledge. Depending on the purpose, the service could return detailed records, summarized knowledge, or knowledge covering a specified time period.

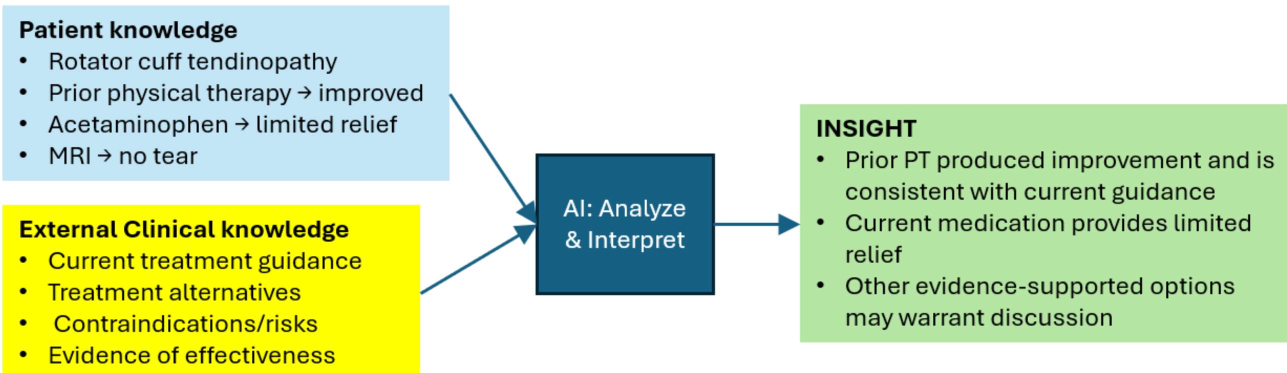


Authentication identifies the individual or system making the request, while privileges determine which knowledge the requester is authorized to receive. The same retrieval service could therefore produce different results depending on the requester and purpose, including identified patient information, de-identified information, or aggregate results. These controls separate access to longitudinal knowledge from the AI that subsequently analyzes it.

Standardized retrieval services also provide AI with more predictable inputs. Instead of requiring each AI application to locate, interpret, and filter an unrestricted collection of records, the retrieval service supplies defined tables of relevant longitudinal knowledge. AI can then concentrate on the next task: analyzing that knowledge to determine what is significant for the purpose for which it was requested.

Analyze & Interpret: Knowledge → Insight

A Longitudinal Patient History may contain decades of knowledge, but its value depends on determining what matters for a particular purpose. Standardized retrieval services provide the authorized patient knowledge appropriate to the task. Analysis may also combine that patient-specific knowledge with clinical guidelines, treatment alternatives, evidence of effectiveness, or other knowledge relevant to the purpose.



AI can then analyze the retrieved knowledge to identify relationships, trends, changes, risks, gaps, or other findings that might otherwise require substantial human effort to discover. For a provider encounter, this could include recognizing that a condition is worsening, identifying a previously ineffective treatment, detecting a potential conflict with another condition or medication, or highlighting information that may warrant further investigation. The objective is not to repeat or summarize everything retrieved, but to identify what is significant enough to deserve attention.

The analysis may also require knowledge beyond the individual patient's history. Evaluating treatment options could combine the patient's conditions and prior treatments with outcomes and costs observed among comparable patients. Prior authorization could compare patient-specific evidence with payer criteria. Other uses may incorporate clinical guidelines, policies, population experience, or other information appropriate to the question being addressed.

What constitutes useful insight therefore depends on the purpose. AI must be taught what to look for, what additional knowledge or rules to apply, and how to distinguish significant findings from information that is merely present. The resulting insight should identify the evidence supporting it and communicate uncertainty where appropriate, allowing the healthcare professional or other authorized user to understand why something was highlighted and determine what action, if any, should be taken.

Making AI Reliable for Healthcare Use

The AI functions described in this architecture would require substantial development and validation before they could be relied upon for healthcare use. A capable Large Language Model is only a starting point. Each function must precisely define what the AI is expected to do, establish how healthcare information and conflicting evidence should be interpreted, provide the knowledge and rules required for the task, and demonstrate through extensive testing that the results are sufficiently accurate and reliable for their intended use. This would require continuing collaboration among clinicians, healthcare organizations, payers, information architects, AI specialists, and other domain experts.

Instructions and definitions can establish the meaning of the information being processed and the expected output. For clinical information extraction, these instructions might define how to distinguish a confirmed diagnosis from a suspected condition, patient-reported history, family history, or a ruled-out condition. They can also specify how extracted information maps to the longitudinal data model and what provenance must be retained.

Representative Cases can teach the AI how to apply those definitions to realistic healthcare information. Representative clinical notes can be paired with the facts and relationships that should be extracted. More difficult examples can demonstrate how to handle ambiguity, conflicting statements, incomplete information, or terminology that varies among providers. Similar examples can be developed for reconciliation, treatment analysis, prior authorization, and other defined functions.

Rules can constrain decisions that should not be left solely to an AI model's judgment. Reconciliation rules might specify how to consider the source, timing, and strength of evidence when information conflicts. Prior-authorization rules can define the criteria that must be satisfied. Other functions may combine deterministic processing with AI interpretation so that established healthcare or business rules are applied consistently.

External knowledge may also be required. The Longitudinal Patient History describes what is known about an individual patient, but some questions require information beyond that history. Treatment analysis might use clinical guidelines and outcomes from comparable patients. Prior authorization might require the payer's criteria for pre-approval of the proposed treatment or procedure. Other applications may require policies, reference information, cost data, or population-level evidence appropriate to the task.

Validation must establish that each AI function performs reliably across different conditions, providers, terminology, documentation styles, patient populations, and unusual or conflicting situations. Testing must include not only common cases, but cases specifically designed to expose incorrect inference, missing information, unsupported conclusions, and other potentially consequential errors.

The objective is therefore not to train a single AI system to "understand healthcare." Instead, it defines specific AI functions within the architecture and provides each with the instructions, examples, rules, knowledge, and validation appropriate to its purpose. As AI models improve, you can replace or upgrade individual models without changing the fundamental information architecture or the responsibilities assigned to each function.

Conclusion

The opportunity for AI in healthcare is broader than giving an AI system a patient's records and asking it to recommend a diagnosis or treatment. Healthcare already generates more information than participants can reasonably find, reconcile, analyze, and understand. A national Longitudinal Patient History makes more information available; AI can help make it usable.

Within the Future Healthcare Information Architecture, AI performs specific functions: extracting clinical information from data, reconciling it into longitudinal knowledge, and analyzing authorized knowledge to produce insight for a particular purpose. These functions can be governed by defined instructions, rules, external knowledge, and validation while preserving the evidence supporting their results.

The objective is not to replace human judgment with an automated answer. It is to use AI to transform **Data → Information → Knowledge → Insight**, giving healthcare professionals and other authorized participants better information for **human decision-making**.

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